



D5 User Evaluation

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D5a - Method for User Evaluation

We selected participants using opportunity sampling from within our Engineering cohort. This is because the game is designed primarily for university students so our selected population seemed representative of our intended users. Each team member recruited one participant meaning we had a total of 6 participants, ages 18 and over, for our evaluation. Following this approach was cost-effective and saved time when organising the sessions for each participant. This approach also aligns with the client's requirement of the game being suitable for young adult users.

Our data collection combined three different methods:

- Questionnaire using google forms: Users complete the form after doing an initial test run of the game. Used to gather feedback on overall user experience.
- Verbal think aloud: Asked participants to describe their thought process while completing the set of tasks created for the task based evaluation.
- Observation notes: Made while the participant completed the tasks. Allowed us to collect behavioural data while the participants interacted with the system.

We created a list of pre-defined tasks that would cover all core interactions with the game based on the user and functional requirements such as FR_ACHIVEMENTS. The tasks were kept simple but goal-oriented to ensure the participants would be able to understand the tasks clearly and to aid us with post feedback development.

The sessions were run in-person to minimise miscommunication. We briefed each participant of the purpose of the evaluation and what they would be participating ensuring to clarify the evaluation was to test the system, not the users ability to use the system. Each session followed the same structure:

- The participant was welcomed and briefed
- The participant completed the tasks under observation
- The participant completed the questionnaire

By keeping all the sessions consistent, it ensured that all evaluation was kept fair and accurate.

All participants completed consent forms and were reminded that they could withdraw from the evaluation any time . Only relevant information was collected, such as age and gender, and all feedback was stored anonymously.

Using multiple data collection methods increased the reliability of the findings by allowing us to compare behaviours, real-time verbal feedback and questionnaire responses.

Six participants were considered sufficient for formative usability testing, as this is the number of members in our group and is accepted to identify the majority of usability issues whilst taking into consideration time and resource demands. This allowed us to identify clear improvements while keeping testing practical within the project timeline.

Although opportunity sampling limits wider generalisation, this limitation was acceptable as the university students are the intended user group for the game.

Our questionnaire responses were summarised in an organised and easy-to-read format, which helped us to prioritise which areas of the game required refinement in future development stages.

Overall, this evaluation approach provided a structured and reliable way to assess the game's usability and alignment with user requirements, utilising questionnaires and verbal feedback.

D5b - Usability Problems

Usability Problems	User severity ratings [1]	Solution
Couldn't access the achievements once the game had finished	4	Achievements should be accessible via the end screen
Setting screen is not implemented	4	Implement setting screen
Didn't realise time is removed when sprinting	3	Have text or animation on screen to show when sprinting
End screen said game over and not congratulations when successful	3	Change text on the screen when the player escapes in time
Not obvious when fell into the basement	2	Use the hole in the floor animation or use of text
Doors missed as not realised they were doors	2	Have welcome signs on the map to make the door more obvious
Basement maze hard to navigate without the torch	2	N/A as the maze is easier to navigate once the torch has been picked up
Game too easy, didn't take long enough to complete and not enough challenge	2	N/A as more events are going to be implemented
Not obvious how to finish the game	2	Add a sign to show exit to the game
Vending machine blends into background, not obvious	2	Make more obvious through use of text that the event is available
Was not aware of picking up the glasses and of their positive impact	2	Have larger text to explain glasses have been used and move the glasses
Didn't understand what the individual achievements mean	2	Add a small explanation below the achievement
Too many instructions to remember	2	Make the instructions accessible from the gameplay screen
Text become blurry when extending the screen	1	Adjust text size when extending screen
Library does not have 'University feel' (66% said it does not represent the University library well or not sure)	1	Add text for visual aid to link to the University.
Background music does not fit the game	0	N/A
Basement was a waste of time when entering the second time	0	N/A

References

[1]J. Nielsen, "Severity Ratings for Usability Problems," *Nielsen Norman Group*, 1994.
<https://www.nngroup.com/articles/how-to-rate-the-severity-of-usability-problems/>